MINI PROJECT PROPOSAL

Development of a Low-Cost Biosensor for the Early Detection of Infectious Diseases

Submitted By: Laiba Ishtiaq
Program: Biotechnology

1. Introduction

A biosensor is a device that consists of biological sensing element,such as enzymes, antibodies, aptamers, or nucleic acid probes with a transducer that detects biochemical events and converts them into electrical, optical, or mechanical signals.Since biosensors can identify disease markers rapidly and accurately without requiring complex laboratory equipment, they are widely used in point-of-care (POC) diagnostic testing (Parihar *et al.*, 2025).

Traditional diagnostic methods, such as enzyme-linked immunosorbent assays (ELISA), polymerase chain reaction (PCR), and culture-based tests, are considered as the gold standard for disease diagnosis due to their high accuracy,but they require specialized equipment,trained professionals,reliable electricity,and cold-chain storage of reagents.These requirements limit their use in rural clinics, resource-limited healthcare facilities, and emergency outbreak situations.Paper-based and electrochemical biosensors provide a useful alternative to traditional diagnostic methods because they are low-cost, easy to operate, disposable, and can be connected to smartphones or portable devices for reading results (Sahraei *et al.,* 2022; Wang et al., 2022).

The purpose of this mini project is to develop and evaluate an affordable biosensor prototype for detecting disease indicators, including COVID-19 antigens, specific microRNAs, or glucose levels in the body (Choi, 2020; Hunt *et al.,* 2024; Hunt & Slaughter, 2025; Das *et al.,* 2025).This project aims to develop a low-cost biosensor by using affordable materials,basic nanotechnology for signal enhancement,making it practical for resource-limited settings.

2. Problem Statement

Accurate and early diagnosis is important for proper treatment and preventing the spread of infections.However, many people in rural and poor areas do not have access to modern laboratory testing facilities. Many diagnostic tests are done in major hospitals or laboratories.Therefore, patient samples have to travel long distances,which can delay diagnosis and affect sample quality. However, current point-of-care diagnostic tools are often too expensive for use in low- and middleincome areas. In addition, they depend on special cartridges or foreign-made parts that are hard to restock in local markets (Parihar *et al.*, 2025).Therefore there is a gap between the need of diagnostic testing methods in resource limited settings and the availability of affordable and appropriate diagnostic technology.

3. Objectives

1. To choose an appropriate biomarker and detection method for designing an affordable biosensor.
2. To fabricate and optimize a low-cost biosensor prototype using affordable materials and simple fabrication techniques.
3. To evaluate the biosensor's sensitivity, accuracy, and affordability potential use in resource-limited settings.

4. Methodology

4.1 Overall Approach

The project will involve step-by step process o designing, fabricationg, testing the biosensor device. A target biomarker an ddetection technique will be selected by reviewing the literature on low-cost biosensors.The prototype will then be made using affordable materials and optimized through laboratory experiments.

4.2 Materials and Sensor Design

The biosensor will be developed using paper-based or screen-printed carbon electrodes as they are affordable, disposable and easy to use. To improve the sensor's sensitivity, the electrode may be coated with affordable nanomaterials, such as carbon nanotubes, gold nanoparticles, or chitosan films, which can enhance its detection performance (Lin et al., 2025).

The sensor surface will be coated with a biological recognition element, such as an antibody, aptamer, or DNA probe that can recognize a specific disease marker by using commonly used chemical attachment methods (Hunt *et al.*, 2024; Hunt & Slaughter, 2025).

4.3 Fabrication Process

1. Design and print the electrode pattern or microfluidic channel on the selected substrate.
2. Modify the working electrode with the chosen nanomaterial to enhance its performance.
3. Immobilise the bio-recognition element on the surface and prevent unwanted binding.
4. Assemble the test strip by adding a reference and counter electrode (for electrochemical sensors) or a colour-producing reagent area (for colorimetric systems).

4.4 Testing and Validation

The fabricated sensor prototype will be analysed using a portable potentiostat (for electrochemical sensors) or a smartphone colour analysis app for colorimetric sensors. Different concentrations of known biomarkers will be used to determine the detection limit and measurement range.The sensor's selectivity will be checked by testing it with similar non-target molecules, and its reliability will be assessed by testing multiple sensor samples.

5. Expected Outcomes

1. A working biosensor device that can detect the target disease biomarker within the required clinical range.
2. An estimated production cost for each sensor, highlighting its potential affordability compared to currently available point-of-care diagnostic tests.
3. A documented fabrication method that can be expanded or modified to detect other disease biomarkers in future studies.

This project is expected to demonstrate that affordable, locally manufactured diagnostic devices can support early disease detection and can be used in healthcare settings with limited resources.

6. References

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